

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 0603 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 Total Marks:25

Code of Conduct:

I X. Joshua 192312188 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

X. Joshua

Annexure A

Short Answer Test

1. Perceptron Classification (Pass/Fail)

Given

- $w_1=0.6$
 $w_1 = 0.6$
- $w_2=0.4$
 $w_2 = 0.4$
- Bias $b=-0.5$
 $b = -0.5$
- $x_1=2$
 $x_1 = 2$ (Study Hours)
- $x_2=1$
 $x_2 = 1$ (Attendance)

Net Input

$$\begin{aligned} y_{in} &= w_1x_1 + w_2x_2 + b \\ y_{in} &= w_1x_1 + w_2x_2 + b \\ &= (0.6 \times 2) + (0.4 \times 1) - 0.5 = (0.6 \times 2) + (0.4 \times 1) - 0.5 = 1.2 + 0.4 - 0.5 = 1.2 + 0.4 - 0.5 = 1.1 \end{aligned}$$

Step Activation Function

If $y_{in} \geq 0$, Output = 1

If $y_{in} < 0$, Output = 0

Since

$$1.1 > 0$$

Output = 1

Final Answer

Student Passes.

2. Delta Rule (Weight Update)

Given

Initial Weight = 0.5

Learning Rate = 0.1

Input = 2

Target = 1

Actual Output = 0.6

Error

$$\text{Error} = \text{Target} - \text{Actual} = 1 - 0.6 = 0.4$$

Weight Update

$$\Delta w = \eta \times \text{Error} \times \text{Input} = 0.1 \times 0.4 \times 2 = 0.08$$

New Weight

$$w_{\text{new}} = 0.5 + 0.08 = 0.58$$

Final Answer

- Error = 0.4
- Weight Update = 0.08
- New Weight = 0.58

3. Confusion Matrix

Given

$$\text{TP} = 180$$

$$\text{TN} = 150$$

$$\text{FP} = 20$$

$$\text{FN} = 50$$

Total

$$= 180 + 150 + 20 + 50 = 400$$

Accuracy

$$\frac{\text{TP} + \text{TN}}{\text{Total}} = \frac{180 + 150}{400} = \frac{330}{400} = 82.5\%$$

Precision

$$\frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{180}{180 + 20} = \frac{180}{200} = 90\%$$

Recall (Sensitivity)

$$= \frac{180}{180+50} = \frac{180}{230} = 78.26\%$$

$$= \frac{180}{230} = 78.26\%$$

Specificity

$$= \frac{150}{150+20} = \frac{150}{170} = 88.24\%$$

$$= \frac{150}{170} = 88.24\%$$

Conclusion

- Accuracy = 82.5%
- Precision = 90%
- Recall = 78.26%
- Specificity = 88.24%

Model is fairly reliable, but recall should be improved since false negatives are important in medical diagnosis.

4. Genetic Algorithm

Fitness Function

$$f(x) = x^2$$

x Fitness

2 4

4 16

6 36

8 64

Selection

Best Individuals

- 8
- 6

Crossover

Binary

$$8 = 1000$$

$$6 = 0110$$

Exchange LSB

Child 1 = 1000 (8)

Child 2 = 0110 (6)

(Depending on the crossover point chosen, offspring may vary. Mention that one valid crossover point is assumed.)

Mutation

Randomly flips one bit.

Example

1000 → 1001

Benefits

- **Maintains diversity**
- **Avoids local optimum**
- **Prevents premature convergence**

5. Training vs Validation Accuracy

Training

60% → 95%

Validation

58% → 72% → 68%

Problem

Overfitting

Reason

Model memorizes training data and cannot generalize to unseen data.

Solutions

- **Early Stopping**
- **Dropout**

(or Regularization, Data Augmentation)

Importance of Validation Accuracy

- Measures performance on unseen data.
- Helps detect overfitting.
- Selects the best model.

6. Perceptron (Loan Approval)

Given

$$w_1=0.7$$

$$w_2=0.5$$

$$\text{Bias}=-0.6$$

$$\text{Income}=1.5$$

$$\text{Credit Score}=2$$

Net Input

$$\begin{aligned} &=(0.7 \times 1.5) + (0.5 \times 2) - 0.6 = (0.7 \times 1.5) + (0.5 \times 2) - 0.6 \\ &= 1.05 + 1 - 0.6 = 1.05 + 1 - 0.6 = 1.05 + 1 - 0.6 = 1.45 = 1.45 = 1.45 \end{aligned}$$

Output

Since

$$1.45 > 0$$

$$\text{Output}=1$$

Final Answer

Loan Approved

7. Delta Rule

Given

$$\text{Weight}=0.4$$

$$\text{Learning Rate}=0.2$$

$$\text{Input}=3$$

$$\text{Target}=0.9$$

Output=0.5

Error

$$=0.9-0.5=0.9-0.5=0.9-0.5=0.4=0.4=0.4$$

Weight Update

$$=0.2 \times 0.4 \times 3 = 0.2 \times 0.4 \times 3 = 0.2 \times 0.4 \times 3 = 0.24 = 0.24 = 0.24$$

New Weight

$$=0.4+0.24=0.4+0.24=0.4+0.24=0.64=0.64=0.64$$

8. Confusion Matrix

TP=120

TN=200

FP=30

FN=50

Total=400

Accuracy

$$= \frac{320}{400} = 0.8$$

=80%

Precision

$$= \frac{120}{150} = 0.8$$

=80%

Recall

$$= \frac{120}{170} \approx 0.7059$$

=70.59%

Specificity

$$= \frac{200}{230} \approx 0.8696$$

=86.96%

F1 Score

$$= 2PRP + R = \frac{2PR}{P+R} = P + R \frac{2PR}{P+R}$$

$$= 2(0.8)(0.7059)0.8 + 0.7059 = \frac{2(0.8)(0.7059)}{0.8 + 0.7059} = 0.8 + 0.7059 \frac{2(0.8)(0.7059)}{0.8 + 0.7059}$$

$$\approx 75\%$$

Conclusion

Recall is only 70.59%, so the model is not ideal when false negatives are critical (e.g., disease detection).

9. Genetic Algorithm

Fitness

$$f(x) = 2x + 3 \quad f(x) = 2x + 3 \quad f(x) = 2x + 3$$

x Fitness

1 5

3 9

5 13

7 17

Selection

Best

-7

-5

Single Point Crossover

Example

7 = 111

5 = 101

After crossover

Children

101

111

Mutation

Flips one bit.

Benefits

- Prevents premature convergence
- Explores new solutions

10. Validation Loss

Training Loss ↓ continuously

Validation Loss ↓ then ↑

Problem

Overfitting

Reason

Model memorizes training data.

Solutions

- Early Stopping
- Regularization (L1/L2)

(or Dropout)

Generalization improves.

11. Perceptron (Purchase Prediction)

Given

$w_1=0.5$

$w_2=0.3$

Bias=-0.4

Advertisement=2

Income=1

Net Input

$$=(0.5 \times 2) + (0.3 \times 1) - 0.4 = (0.5 \times 2) + (0.3 \times 1) - 0.4 = 1 + 0.3 - 0.4 = 1 + 0.3 - 0.4 = 0.9 = 0.9 = 0.9$$

Output

Since

$$0.9 > 0$$

$$\text{Output} = 1$$

Final Answer

Customer Purchases Product

12. Delta Rule

Given

$$\text{Weight} = 0.6$$

$$\text{Learning Rate} = 0.15$$

$$\text{Input} = 2.5$$

$$\text{Output} = 0.7$$

$$\text{Target} = 1$$

Error

$$= 1 - 0.7 = 1 - 0.7 = 1 - 0.7 = 0.3 = 0.3 = 0.3$$

Weight Update

$$= 0.15 \times 0.3 \times 2.5 = 0.15 \times 0.3 \times 2.5 = 0.15 \times 0.3 \times 2.5 = 0.1125 = 0.1125 = 0.1125$$

New Weight

$$= 0.6 + 0.1125 = 0.6 + 0.1125 = 0.6 + 0.1125 = 0.7125 = 0.7125 = 0.7125$$

13. Confusion Matrix

$$\text{TP} = 140$$

$$\text{TN} = 180$$

$$\text{FP} = 25$$

$$\text{FN} = 35$$

$$\text{Total} = 380$$

Accuracy

$$= \frac{320}{380} = \frac{320}{380} = 0.8421$$

$$= 84.21\%$$

Precision

$$= \frac{140}{165} = 84.85\%$$

$$= 84.85\%$$

Recall

$$= \frac{140}{175} = 80\%$$

$$= 80\%$$

Specificity

$$= \frac{180}{205} = 87.80\%$$

$$= 87.80\%$$

F1 Score

$$\approx 82.4\%$$

Conclusion

Since precision is high (84.85%), the model is reasonably suitable for fraud detection, where reducing false positives is important.

14. Genetic Algorithm

Fitness

$$f(x) = 3x + 2$$

x Fitness

2 8

5 17

7 23

9 29

Selection

Best

-9

-7

Crossover

Example

$$9 = 1001$$

7=0111

Single-point crossover → Generate offspring by exchanging the bits after a chosen crossover point (offspring depend on the point selected).

Mutation

Random bit flip.

Benefits

- **Maintains diversity**
- **Avoids local optima**
- **Finds better solutions**

15. Training vs Validation Accuracy

Training

65%→97%

Validation

63%→75%→72%

Problem

Overfitting

Reason

Model memorizes training data instead of learning general patterns.

Solutions

- **Dropout**
- **Data Augmentation**

(also Early Stopping and Regularization)

Importance of Validation Metrics

- **Evaluate performance on unseen data.**
- **Detect overfitting.**
- **Help choose the best model.**
- **Ensure good generalization.**